

E-commerce Revenue Leakage & Risk Prediction

End-to-End Data Analytics & Machine Learning Project

BY SHRUTI VERMA

📁 Excel | Power BI | Python | Scikit-learn | Streamlit



Made with GAMMA

Project Overview

This project analyzes e-commerce order data to identify risky transactions and revenue leakage patterns — combining analytics, dashboards, ML, and deployment to solve a real-world business problem.



Excel Analysis



Power BI Dashboard



Python EDA



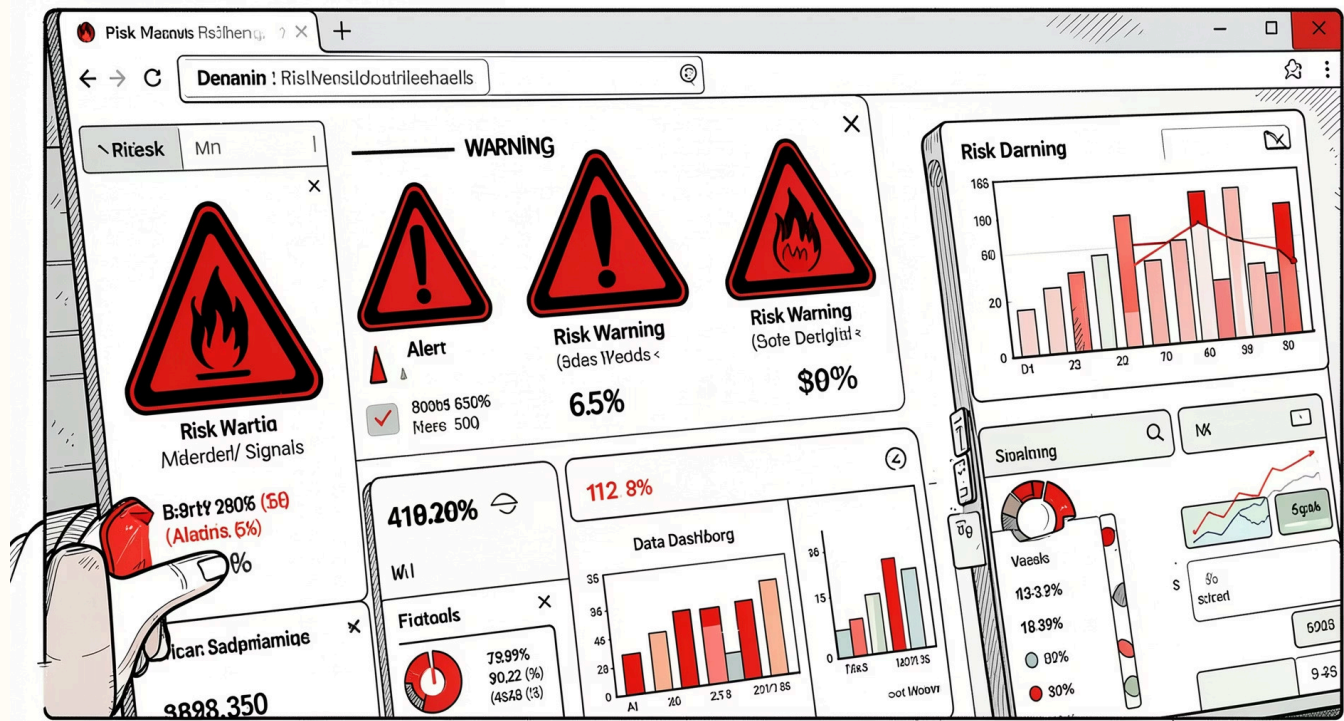
ML Model



Streamlit Deployment

Problem Statement

E-commerce businesses face order cancellations, payment risks, low reviews, and revenue leakage. This project addresses these challenges through a structured analytics and ML workflow.



Analyze Behavior

Customer & payment patterns

Identify Risk

Flag risky orders

Predict

Build a risk prediction system

Visualize

Create actionable dashboards

Tools & Technologies

Technologies Used

Excel

Power BI

Python

Pandas & NumPy

Matplotlib

Scikit-learn

Streamlit

Key Concepts Applied

- Data Cleaning & Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Machine Learning
- Model Deployment



Insert tools collage or tech stack screenshot here

Excel Analysis

Data Cleaning

Revenue KPIs

Review Score

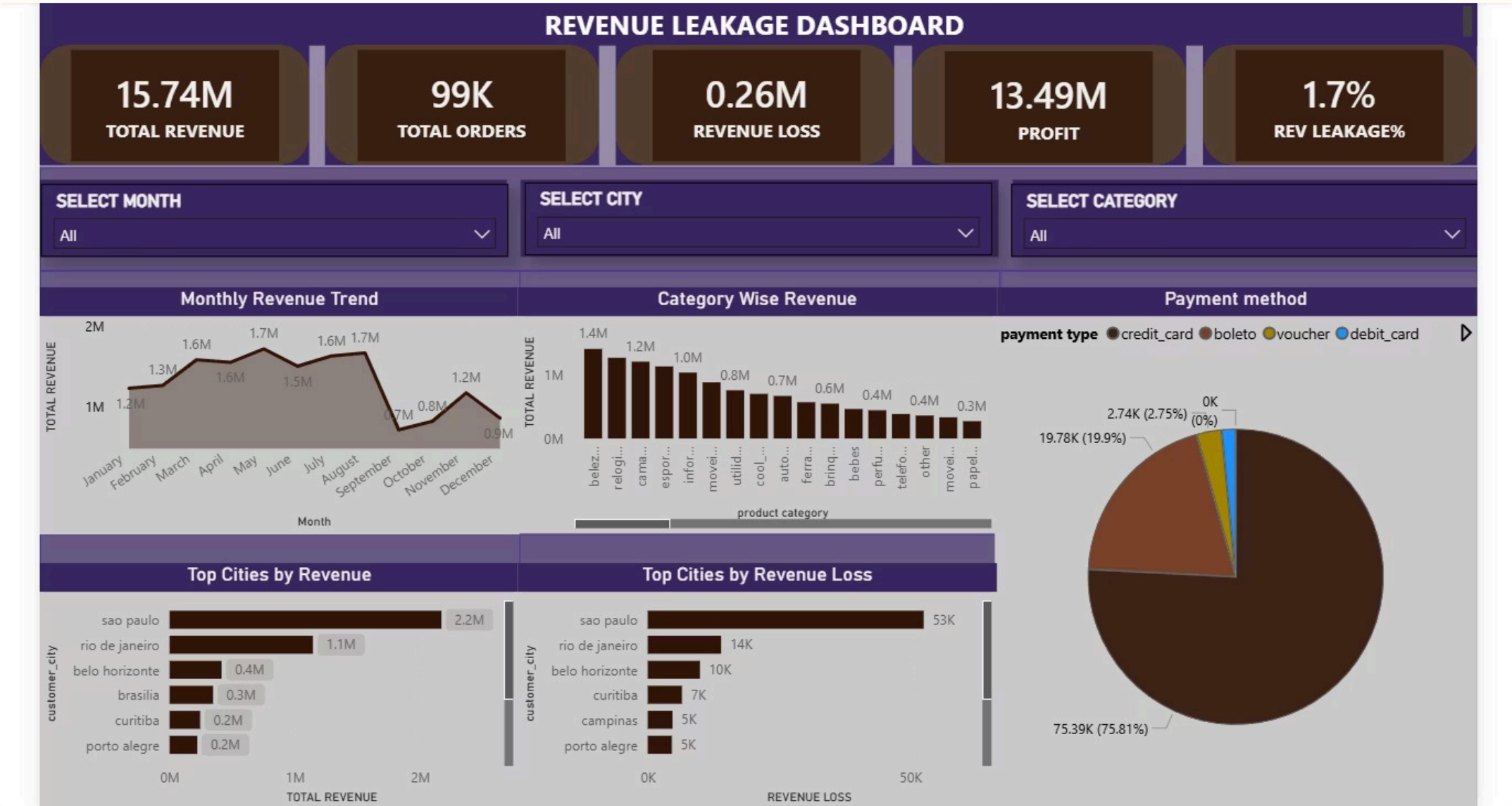
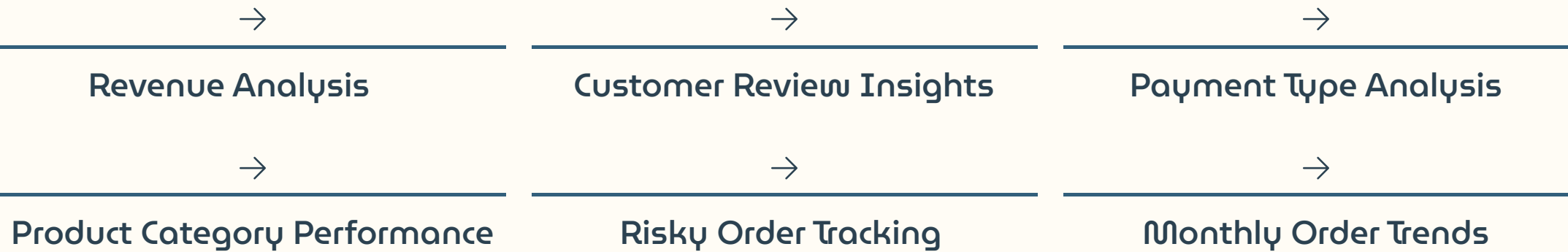
Payment Trends



Made with **Gamma**

Power BI Dashboard

Dashboard Features



Python EDA & Feature Engineering

```
Aug    413375.65
Dec    200163.58
Feb    305480.25
Jan    302486.83
Jul    391387.44
Jun    358756.47
Mar    372034.64
May    440273.17
Nov    285002.13
Oct    193906.97
Sep    173035.49
Name: payment value, dtype: float64
```

```
monthly_revenue.plot(kind="line")
plt.title("MONTHLY REVENUE")
plt.xlabel("CITY")
plt.ylabel("REVENUE")
plt.show()
```

```
print("TOTAL_REVENUE = ",total_revenue)
```

```
TOTAL_REVENUE = 3788643.57
```

Total Orders

```
total_orders = data.shape[0]
print("TOTAL_ORDERS = ",total_orders)
```

```
TOTAL_ORDERS = 23711
```

Shows how much a customer spends per order?

```
average_order_value = data['payment value'].mean()
print("AVERAGE ORDER VALUE = ",average_order_value)
```

```
AVERAGE ORDER VALUE = 159.78421703007044
```

Which cities give most money?

EDA Tasks

- Data preprocessing & null value checks
- Data type conversion
- Revenue, review & payment behavior analysis
- Visualization using Matplotlib

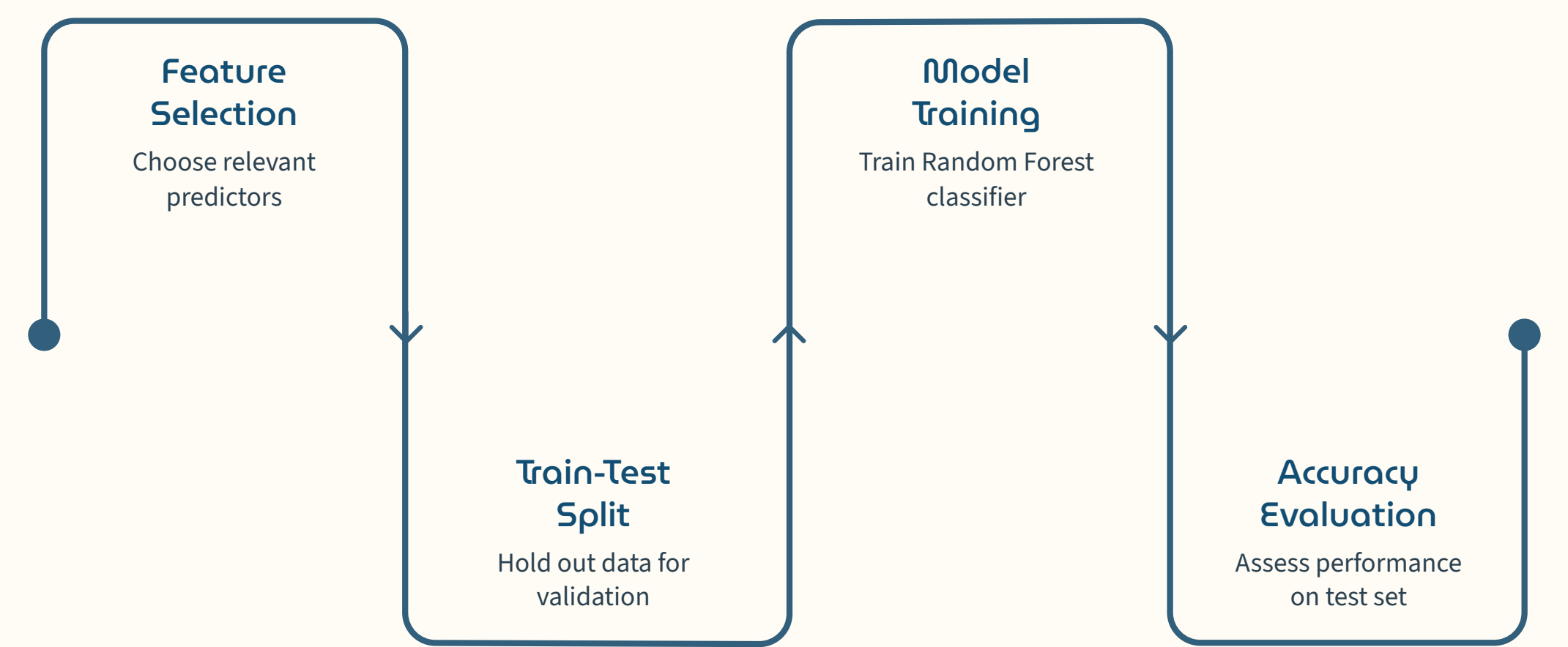
Risk Flag Creation

Business risk rules were combined into a target column **"Risk Flag"** using:

- Bad customer reviews
- Payment-related risks
- Low order value behavior
- Cancellation/return patterns

Machine Learning Model

A **Random Forest Classifier** was trained to predict whether an order is risky or safe.



Why Random Forest? Handles mixed data well, reduces overfitting, and delivers strong classification accuracy.

Handles Mixed Data	Reduces Overfitting	High Accuracy
--------------------	---------------------	---------------

Model Evaluation

Accuracy = Correct Predictions / Total Predictions

```
model_accuracy_score=accuracy_score(y_test,pred)
print("Accuracy score is :",model_accuracy_score)
```

Accuracy score is : 0.9987349778621126

Confusion Matrix

```
confusion_matrix(y_test,pred)
```

array([[2743, 1],
 [5, 1994]])

Classification Report

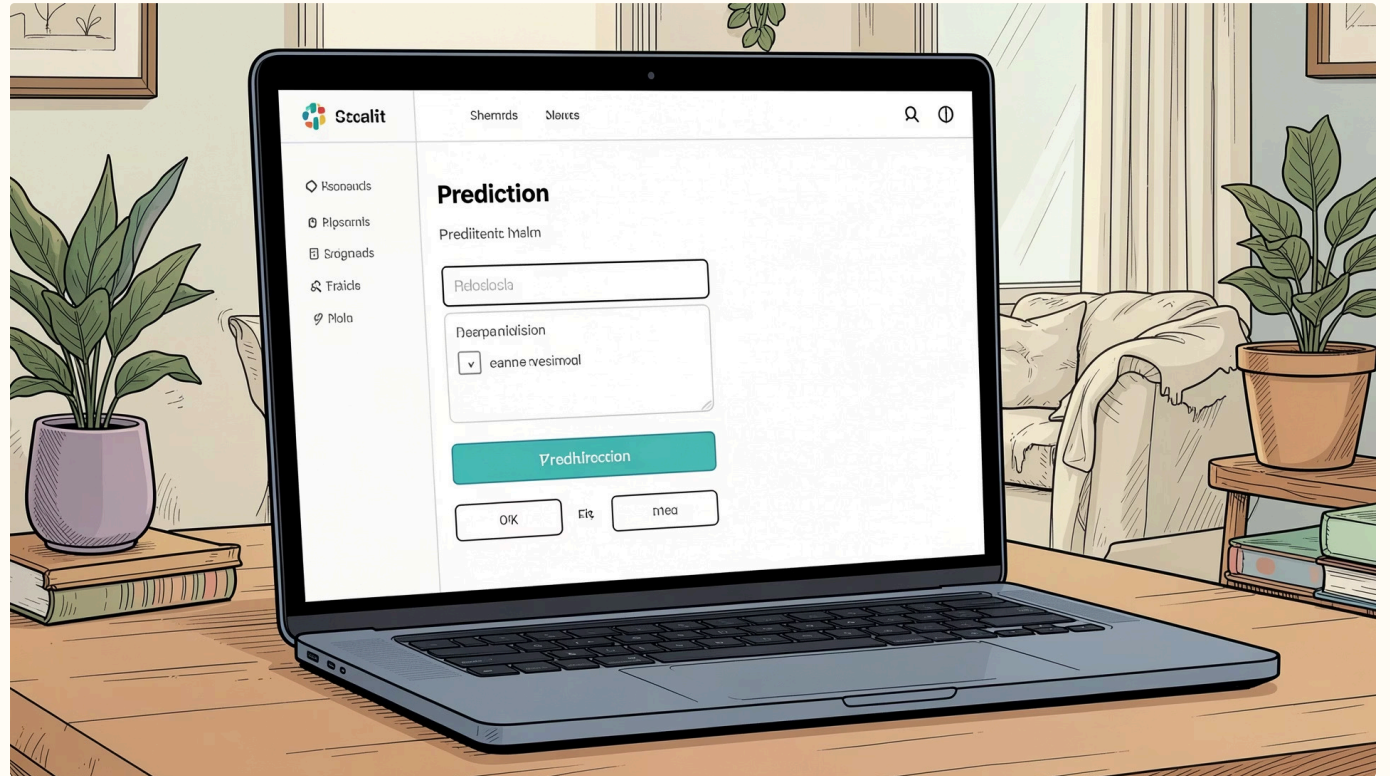
```
print(classification_report(y_test,pred))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	2744
1	1.00	1.00	1.00	1999
accuracy			1.00	4743
macro avg	1.00	1.00	1.00	4743
weighted avg	1.00	1.00	1.00	4743

Streamlit Deployment

How It Works

The trained model was saved using **Pickle** and deployed as an interactive web app via Streamlit.



Enter Order Details

Select Payment & Product
Info

Predict: Risky or Safe

Edit All Ctrl + E

Edit all objects in the file. Preview the file by deselecting all.



E-commerce Risk Predictor

Enter order details below

Payment Amount

1000

Review Score

3

Customer City

Mumbai

Payment Type

credit_card

Product Category

electronics

Month

January

Predict Risk



Risky Order

Results & Conclusion

End-to-End Workflow

Understood the complete data science pipeline from analysis to deployment

Business Dashboards

Built Power BI dashboards for actionable insights

ML & Deployment

Trained a Random Forest model and deployed it as a working web app

This project provided practical hands-on experience in **Data Analytics** and **Data Science** — combining analytics, ML, and deployment to solve real business problems.